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Experiment Name : _CONTROL STATEMENTS,TERINARY,ARRAY, OBJECT OF ARRAY ___

Date : ___21/02/2026_____ */

Programs using Condition

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for/while/Terinary:

1. Find factorial of a number.

2.Reverse a number.

3.Check if a number is even or odd using ternary.

4. Write a Java program to check whether a given integer is a palindrome number.

```
import java.util.Scanner;


public class Factorial {
   public static void main(String[] args) {


      Scanner sc = new Scanner(System.in);


      System.out.print("Enter number: ");
      int n = sc.nextInt();


      int fact = 1;


      for (int i = 1; i <= n; i++) {
```

```
        fact *= i;
    }

    System.out.println("Factorial = " + fact);
  }
}
```

```
C:\Users\Sharayu\OneDrive\Documents\JavaPractical>javac Factorial.java

C:\Users\Sharayu\OneDrive\Documents\JavaPractical>java Factorial
Enter number: 4
Factorial = 24
```

```
import java.util.Scanner;

public class ReverseNumber {
  public static void main(String[] args) {

    Scanner sc = new Scanner(System.in);

    System.out.print("Enter number: ");
    int num = sc.nextInt();

    int reverse = 0;

    while (num != 0) {
        int digit = num % 10;
        reverse = reverse * 10 + digit;
        num = num / 10;
    }

    System.out.println("Reversed Number = " + reverse);
```

```
    }
}
```

```
C:\Users\Sharayu\OneDrive\Documents\JavaPractical>javac ReverseNumber.java

C:\Users\Sharayu\OneDrive\Documents\JavaPractical>java ReverseNumber
Enter number: 2356
Reversed Number = 6532
```

```
import java.util.Scanner;

public class EvenOddTernary {

    public static void main(String[] args) {


        Scanner sc = new Scanner(System.in);

        System.out.print("Enter a number: ");

        int n = sc.nextInt();


        String result = (n % 2 == 0) ? "Even" : "Odd";


        System.out.println(result);

    }

}
```

```
C:\Users\Sharayu\OneDrive\Documents\JavaPractical>javac EvenOddTernary.java

C:\Users\Sharayu\OneDrive\Documents\JavaPractical>java EvenOddTernary
Enter a number: 26
Even
```

```
import java.util.Scanner;


public class PalindromeNumber {

    public static void main(String[] args) {


        Scanner sc = new Scanner(System.in);
```

```
        System.out.print("Enter a number: ");
        int num = sc.nextInt();

        int original = num;
        int reverse = 0;

        while (num != 0) {
            int digit = num % 10;
            reverse = reverse * 10 + digit;
            num = num / 10;
        }

        if (original == reverse)
            System.out.println("Palindrome Number");
        else
            System.out.println("Not a Palindrome Number");
    }
}
```

```
C:\Users\Sharayu\OneDrive\Documents\JavaPractical>javac PalindromeNumber.java

C:\Users\Sharayu\OneDrive\Documents\JavaPractical>java PalindromeNumber
Enter a number: 2334
Not a Palindrome Number
```

Switch Programs:

1. Write a Java program that performs basic arithmetic operations (+, -, *, /) using a switch statement.

```
import java.util.Scanner;

public class CalculatorSwitch {
```

```
public static void main(String[] args) {

    Scanner sc = new Scanner(System.in);

    System.out.print("Enter first number: ");
    double a = sc.nextDouble();

    System.out.print("Enter operator (+, -, *, /): ");
    char op = sc.next().charAt(0);

    System.out.print("Enter second number: ");
    double b = sc.nextDouble();

    switch (op) {
        case '+':
            System.out.println("Result = " + (a + b));
            break;
        case '-':
            System.out.println("Result = " + (a - b));
            break;
        case '*':
            System.out.println("Result = " + (a * b));
            break;
        case '/':
            if (b != 0)
                System.out.println("Result = " + (a / b));
            else
                System.out.println("Division by zero not allowed");
```

```
            break;
        default:
            System.out.println("Invalid Operator");
    }
  }
}
```

```
C:\Users\Sharayu\OneDrive\Documents\JavaPractical>javac CalculatorSwitch.java

C:\Users\Sharayu\OneDrive\Documents\JavaPractical>java CalculatorSwitch
Enter first number: 12
Enter operator (+, -, *, /): +
Enter second number: 34
Result = 46.0
```

2.Write a Java program that takes a number (1–7) and prints the corresponding day of the week using a switch statement.

```
import java.util.Scanner;


public class DayOfWeek {
  public static void main(String[] args) {


    Scanner sc = new Scanner(System.in);
    System.out.print("Enter number (1-7): ");
    int day = sc.nextInt();


    switch (day) {
      case 1: System.out.println("Monday"); break;
      case 2: System.out.println("Tuesday"); break;
      case 3: System.out.println("Wednesday"); break;
      case 4: System.out.println("Thursday"); break;
      case 5: System.out.println("Friday"); break;
      case 6: System.out.println("Saturday"); break;
      case 7: System.out.println("Sunday"); break;
```

```
        default: System.out.println("Invalid Input");
      }
    }
}
```

```
C:\Users\Sharayu\OneDrive\Documents\JavaPractical>javac DayOfWeek.java

C:\Users\Sharayu\OneDrive\Documents\JavaPractical>java DayOfWeek
Enter number (1-7): 6
Saturday
```

Array Programs

1.Write a Java program that takes n integers as input from the user using a Scanner, stores them in an array, and calculates the sum of all elements.

```
import java.util.Scanner;


public class ArraySum {
   public static void main(String[] args) {


      Scanner sc = new Scanner(System.in);


      System.out.print("Enter number of elements: ");
      int n = sc.nextInt();


      int arr[] = new int[n];
      int sum = 0;


      System.out.println("Enter elements:");


      for (int i = 0; i < n; i++) {
         arr[i] = sc.nextInt();
```

```
        sum += arr[i];
    }

    System.out.println("Sum = " + sum);
  }
}
```

```
C:\Users\Sharayu\OneDrive\Documents\JavaPractical>java ArraySum
Enter number of elements: 5
Enter elements:
2
3
4
5
6
Sum = 20
```

2.Write a Java program that reads n integers into an array and finds the largest and smallest element.

```
import java.util.Scanner;

public class ArrayMaxMin {
  public static void main(String[] args) {

    Scanner sc = new Scanner(System.in);

    System.out.print("Enter number of elements: ");
    int n = sc.nextInt();

    int arr[] = new int[n];

    System.out.println("Enter elements:");
    for (int i = 0; i < n; i++) {
        arr[i] = sc.nextInt();
```

```
        }

        int max = arr[0];
        int min = arr[0];

        for (int i = 1; i < n; i++) {
            if (arr[i] > max)
                max = arr[i];
            if (arr[i] < min)
                min = arr[i];
        }

        System.out.println("Largest = " + max);
        System.out.println("Smallest = " + min);
    }
}
```

```
C:\Users\Sharayu\OneDrive\Documents\JavaPractical>javac ArrayMaxMin.java

C:\Users\Sharayu\OneDrive\Documents\JavaPractical>java ArrayMaxMin
Enter number of elements: 5
Enter elements:
2
4
1
6
4
Largest = 6
Smallest = 1
```

3. Write a Java program to perform addition of two matrices (2D arrays). Input: Two 3×3 matrices entered by the user. Output: A new matrix that is the sum of the two.

```
import java.util.Scanner;
```

```
public class MatrixAddition {
  public static void main(String[] args) {

    Scanner sc = new Scanner(System.in);

    int a[][] = new int[3][3];
    int b[][] = new int[3][3];
    int sum[][] = new int[3][3];

    System.out.println("Enter first matrix:");
    for (int i = 0; i < 3; i++)
      for (int j = 0; j < 3; j++)
        a[i][j] = sc.nextInt();

    System.out.println("Enter second matrix:");
    for (int i = 0; i < 3; i++)
      for (int j = 0; j < 3; j++)
        b[i][j] = sc.nextInt();

    for (int i = 0; i < 3; i++)
      for (int j = 0; j < 3; j++)
        sum[i][j] = a[i][j] + b[i][j];

    System.out.println("Sum Matrix:");
    for (int i = 0; i < 3; i++) {
      for (int j = 0; j < 3; j++)
        System.out.print(sum[i][j] + " ");
      System.out.println();
```

```
    }
  }
}
```

```
C:\Users\Sharayu\OneDrive\Documents\JavaPractical>javac MatrixAddition.java

C:\Users\Sharayu\OneDrive\Documents\JavaPractical>java MatrixAddition
Enter first matrix:
1 2 3
4 5 6
1 6 7
Enter second matrix:
3 5 8
9 0 2
1 3 4
Sum Matrix:
4 7 11
13 5 8
2 9 11
```

4. Matrix Multiplication (2D Array) Problem Statement: Write a Java program to perform multiplication of two matrices. Input: Two matrices (say 2×3 and 3×2). Output: Resultant matrix after multiplication

```
import java.util.Scanner;


public class MatrixMultiplication {
  public static void main(String[] args) {


    Scanner sc = new Scanner(System.in);


    int a[][] = new int[2][3];
    int b[][] = new int[3][2];
    int result[][] = new int[2][2];


    System.out.println("Enter first matrix (2x3):");
    for (int i = 0; i < 2; i++)
      for (int j = 0; j < 3; j++)
```

```
            a[i][j] = sc.nextInt();

        System.out.println("Enter second matrix (3x2):");
        for (int i = 0; i < 3; i++)
            for (int j = 0; j < 2; j++)
                b[i][j] = sc.nextInt();

        for (int i = 0; i < 2; i++) {
            for (int j = 0; j < 2; j++) {
                result[i][j] = 0;
                for (int k = 0; k < 3; k++) {
                    result[i][j] += a[i][k] * b[k][j];
                }
            }
        }

        System.out.println("Result Matrix:");
        for (int i = 0; i < 2; i++) {
            for (int j = 0; j < 2; j++)
                System.out.print(result[i][j] + " ");
            System.out.println();
        }
    }
}
```

```
C:\Users\Sharayu\OneDrive\Documents\JavaPractical>javac MatrixMultiplication.java

C:\Users\Sharayu\OneDrive\Documents\JavaPractical>java MatrixMultiplication
Enter first matrix (2x3):
2 3 4
4 5 6
Enter second matrix (3x2):
2 3 7
1 6 7
Result Matrix:
49 37
79 59
```

. ARRAY OBJECTS:

1. Write a Java program to create a Student class with attributes rollNo and name. Take input for multiple students using Scanner, store them in an array of objects, and then display their details.

```
import java.util.Scanner;

class Student {
  int rollNo;
  String name;

  void input(Scanner sc) {
    System.out.print("Enter Roll No: ");
    rollNo = sc.nextInt();
    sc.nextLine();
    System.out.print("Enter Name: ");
    name = sc.nextLine();
  }

  void display() {
    System.out.println("Roll No: " + rollNo + ", Name: " + name);
  }
}

public class StudentArray {
  public static void main(String[] args) {

    Scanner sc = new Scanner(System.in);

    System.out.print("Enter number of students: ");
    int n = sc.nextInt();
```

```
        Student students[] = new Student[n];

        for (int i = 0; i < n; i++) {
            students[i] = new Student();
            students[i].input(sc);
        }

        System.out.println("Student Details:");
        for (int i = 0; i < n; i++) {
            students[i].display();
        }
    }
}
```

```
C:\Users\Sharayu\OneDrive\Documents\JavaPractical>javac StudentArray.java

C:\Users\Sharayu\OneDrive\Documents\JavaPractical>java StudentArray
Enter number of students: 5
Enter Roll No: 20
Enter Name: sharayu
Enter Roll No: 12
Enter Name: swati
Enter Roll No: 15
Enter Name: shreya
Enter Roll No: 10
Enter Name: sonu
Enter Roll No: 25
Enter Name: monu
Student Details:
Roll No: 20, Name: sharayu
Roll No: 12, Name: swati
Roll No: 15, Name: shreya
Roll No: 10, Name: sonu
Roll No: 25, Name: monu
```

2 Write a Java program to create a Product class with attributes id, name, and price. Store multiple products in an array of objects, then calculate the total inventory value (sum of all product prices).

```
import java.util.Scanner;


class Product {
    int id;
    String name;
```

```
    double price;

    void input(Scanner sc) {
        System.out.print("Enter Product ID: ");
        id = sc.nextInt();
        sc.nextLine();
        System.out.print("Enter Product Name: ");
        name = sc.nextLine();
        System.out.print("Enter Price: ");
        price = sc.nextDouble();
    }
}

public class ProductArray {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter number of products: ");
        int n = sc.nextInt();

        Product products[] = new Product[n];
        double total = 0;

        for (int i = 0; i < n; i++) {
            products[i] = new Product();
            products[i].input(sc);
            total += products[i].price;
```

```
    }

    System.out.println("Total Inventory Value = " + total);
  }
}
```

```
C:\Users\Sharayu\OneDrive\Documents\JavaPractical>javac ProductArray.java

C:\Users\Sharayu\OneDrive\Documents\JavaPractical>java ProductArray
Enter number of products: 3
Enter Product ID: 123
Enter Product Name: car
Enter Price: 123576
Enter Product ID: 687
Enter Product Name: soap
Enter Price: 244
Enter Product ID: 12345
Enter Product Name: bike
Enter Price: 233547
Total Inventory Value = 357367.0
```